

Yanru Chen

Computer Architecture and Performance Modeling

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UCSD ECE Ph.D. candidate focused on memory-centric computer architecture and hardware/software co-design. Research spans heterogeneous accelerators, graph analytics and LLM inference, with SK hynix experience in event-driven simulation and trace replay. Anticipated graduation June 2027.

Technical Focus

Python, C++, MATLAB, Verilog | Performance modeling, event-driven simulation, trace replay, PIM, heterogeneous memory, workload mapping

Industry Experience

SK hynix | San Jose, CA

Jun - Sep 2026

Emerging Memory AI Infrastructure Intern

- Built event-driven simulation and trace-replay workflows to evaluate heterogeneous AI architectures.
- Analyzed compute, memory and data-movement interactions to inform hardware/software co-design.

Research Experience

UC San Diego | Systems Energy Efficiency Lab

Feb 2025 - Present

Graduate Researcher | Advisor: Tajana Rosing

CHIME | DATE 2026 | Shared first author

- Co-developed a chiplet-based near-memory architecture for edge multimodal LLM inference, pairing M3D DRAM and RRAM with memory-aware operator mapping and KV-cache placement.

RAPID-Graph | DATE 2026 | Shared first author

- Co-developed a PIM approach to exact all-pairs shortest paths, using recursive graph partitioning and in-place Floyd-Warshall and min-plus computation in PCM arrays.

HDDB | DAC 2026 | Shared first author

- Co-developed hyperdimensional encodings and in-storage SQL predicate processing on ferroelectric NAND flash; studied data movement and noise tolerance on TPC-DS fact-table workloads.

GEN-Graph | TCAD submission | First author

- Co-developed a heterogeneous PIM architecture with matrix and traversal tiles for dense graph computation and irregular sequence-to-graph alignment; studied data mapping and scheduling.

Education

University of California San Diego

Sep 2024 - Present

Ph.D. candidate, Electrical and Computer Engineering | Anticipated graduation June 2027

Advisor: Tajana Rosing | GPA 3.6/4.0 | Qualifying exam passed March 2026

Tsinghua University

Aug 2021 - Jun 2024

M.S., Electronic Information (Intelligent Manufacturing) | GPA 3.78/4.0

Jilin University

Aug 2017 - Jun 2021

B.E., Electronic Science and Technology | College Honors | GPA 3.53/4.0 | Rank 6/108

Additional Experience and Honors

Earlier research: flexible electronics and semiconductor sensors at UCSD and Tsinghua; CT perfusion analysis and machine learning at Western University.

UCSD CSE Best Doctoral Research Award (2026); DAC Young Fellow (2026); Jacobs Fellowship (2024).